2012/13

Mid semester Examination

KUMASI

March 2013

ME 454 Hydraulics and Pneumatics

Answer All Questions on the question paper.

Time: 45 Minutes

1. In a hydraulic and pneumatic systems, there are five major parts. In the spaces provided, name them and state one function each.

a. Power unit. (1)

Function: Provide the power to operate the system. (1)

b. actuator. (1)

Function: Convert fluid energy into mechanical energy (1)

c. control valve (1)

Function: Direct the flow of the fluid, control the flow rate, pressure or direction (1)

e. Reservoir (1)

Function: Store the oil of the system, and provide the hydraulic fluid to the syste

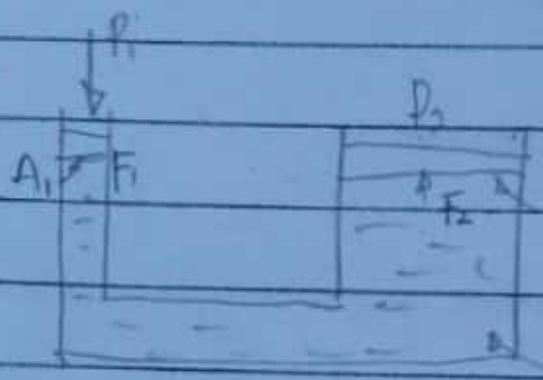
f. Fluid conditioning (1)

Function: keep the hydraulic fluid clean. (1)

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2. State the basic principle by which fluid power systems operate. With simple diagram and some parameters, illustrate force multiplication of a hydraulic system.

Pascal's Principle :- Pressure is transmitted undiminished in an enclosed static fluid. 2



2

~~hydraulic fluid~~ static hydraulic fluid

from principle, $P_1 = P_2$ 1

$F_1/A_1 = F_2/A_2$

$F_2 = F_1 \cdot \left(\frac{A_2}{A_1}\right)$ 1

$A_2 > A_1$ so $F_2 > F_1$ 1 (Force is multiplied by the ratio of the areas)

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December, 2013 Time: 2 hours

ID 6793711 Centre ACCRA

Answer **All** Questions.

Question 1

(a) Define Fluid Power and state kinds of such a system with brief explanations.

(b) State three advantages and three disadvantages each of the systems identified in (

(c) With a simple schematic diagram, illustrate structure of systems identified in (a)

indicating the operational difference between the two kinds.

[10 Marks]

Question 2

(a) State the basic principle by which fluid power systems operate.

(b) One advantage of fluid power system is force multiplication. With simple diagra
illustrate force multiplication of a hydraulic system.

(c) State five safety measures necessary in working with Fluid Power systems.

[10 Marks]

Question 3

(a) State a function each of the following as used in Fluid Power systems.

Check valve, Pressure reducing valve, Fluid, Accumulator, Reservoir, Actuator

booster

(b) Explain the need of schematic/symbolic drawing in hydraulic system.

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(c) In a hydraulic circuit diagram, a label of a component is indicated as **2-3V5.**

Explain the meaning of the elements of the number.

(d) With a simple circuit diagram illustration, explain what an intensifier circuit is.

[20 marks]

Question 4

(a) Figure 1 is a drawing of hydraulic circuit with 4/3 directional control valve in Views A, B and C. Using colour coding, apply the appropriate colour to the various portions of the circuit (position of the valve is indicated with shading) . Indicate with arrow the direction of the piston in each case.

(b) Produce a circuit diagram using standard symbol for the View B of Figure 1.

(c) A hydraulic Engineer designs a hydraulic system. He decided to make use of the following components: double acting cylinder, a booster, an electric motor, hydraulic pump, 4/3 directional control valve, 2/2 directional control valve (normally closed), an accumulator, pressure relief valve, a reservoir and various hose connectors.

Produce a possible circuit diagram using standard symbols with the chosen components.

[30 marks]

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5. Figure 2 is a hydraulic circuit diagram, name the various components as per label provided at respective spaces.

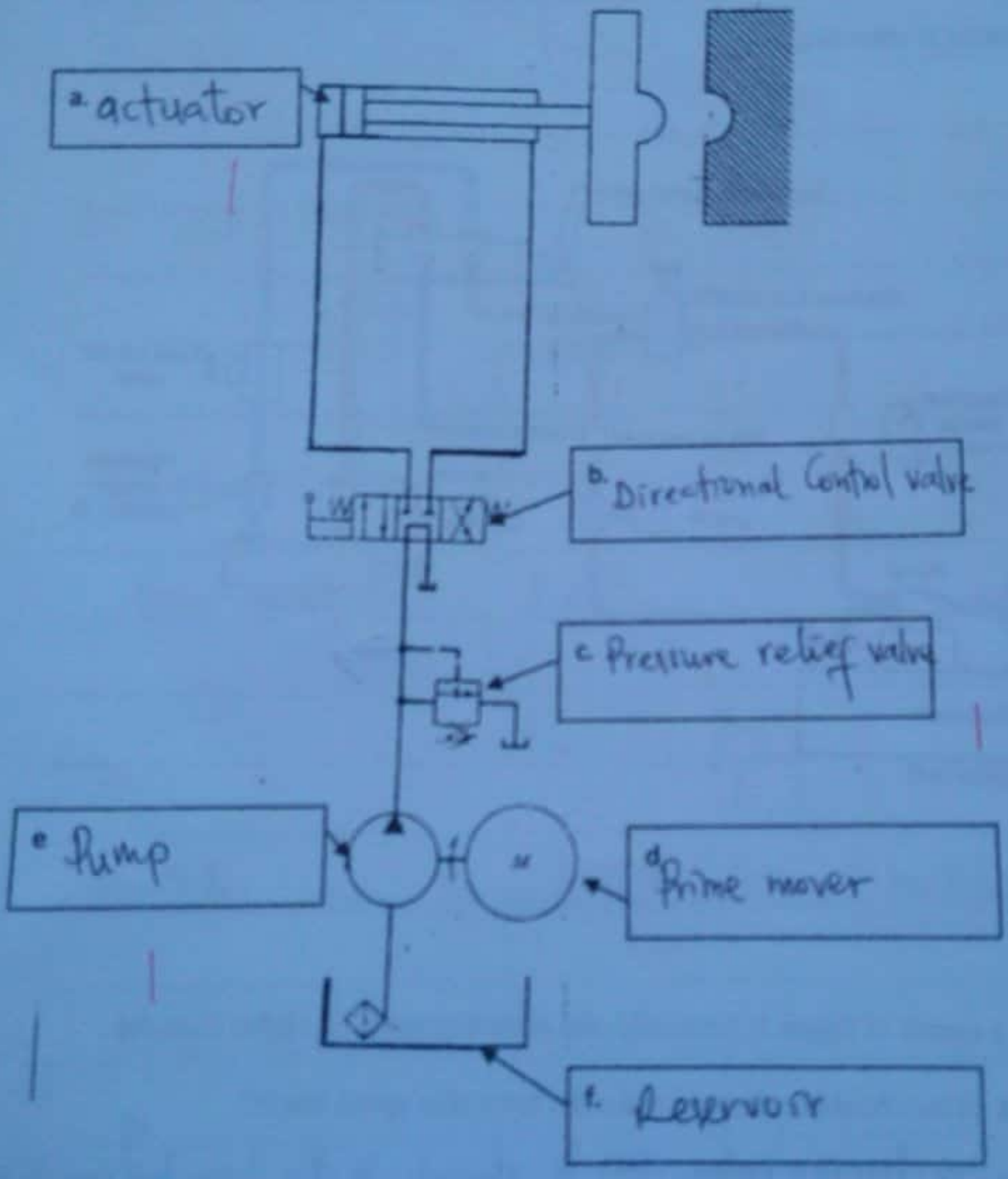


Figure 2.

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KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI
INSTITUTE OF DISTANT LEANING

End of First Semester 2011/12 Examination

B.Sc. Mechanical Engineering (IDL)

ME 453 Hydraulics and Pneumatics

December, 2012 **Time: 2 hours**

Answer **All** Questions in the answer booklet provided.

Question 1

(a) Define Fluid Power and state types of such a system with brief explanations.

(b) State three advantages and three disadvantages each of the systems identified in (a) above.

(c) State five safety measures necessary in working with Fluid Power systems.

[10 Marks]

Question 2

(a) State the basic principle by which fluid power systems operate.

(b) With simple diagram and some parameters, illustrate force multiplication of a hydraulic system.

(c) Illustrate how flow rate and cylinder area of a hydraulic systems are related.

[10 Marks]

Question 3

(a) State a function each of the following as used in Fluid Power systems.

Directional control Valve, Pressure relief valve, Fluid, Accumulator, Reservoir, Actuator and Intensifier

(b) Explain the need of schematic/symbolic drawing in hydraulic system.

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(c) In a hydraulic circuit diagram, a label of a component is indicated as **2-3V5.**

Explain the meaning of the elements of the number.

(d) With a simple circuit diagram illustration, explain what an intensifier circuit is.

[20 marks]

Question 4

(a) Figure 1 is a drawing of hydraulic circuit with 4/3 directional control valve in Views A, B and C. Using colour coding, apply the appropriate colour to the various portions of the circuit (position of the valve is indicated with shading) . Indicate with arrow the direction of the piston in each case.

(b) Produce a circuit diagram using standard symbol for the View B of Figure 1.

(c) A hydraulic Engineer designs a hydraulic system. He decided to make use of the following components: double acting cylinder, a booster, an electric motor, hydraulic pump, 4/3 directional control valve, 2/2 directional control valve (normally closed), an accumulator, pressure relief valve, a reservoir and various hose connectors.

Produce a possible circuit diagram using standard symbols with the chosen components.

[30 marks]

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(c) In a normal cylinder operation, actuating and returning speeds are different. State the different between them and illustrate with a circuit diagram how to make both speeds equal.

(d) With a simple circuit diagram illustration, explain what an intensifier circuit is.

[15 Marks]

Question 4

Answer this question on the question paper.

Figure 1 is hydraulic circuit with 4/3 directional control valve with two of the positions indicated as A and B. State the kind of circuit system. At position A (Figure 1A), indicate with arrows the direction of the cylinders, if actuations are not at the same time provide the order. Do same for position B (Figure 1B).

Regenerative Circuit

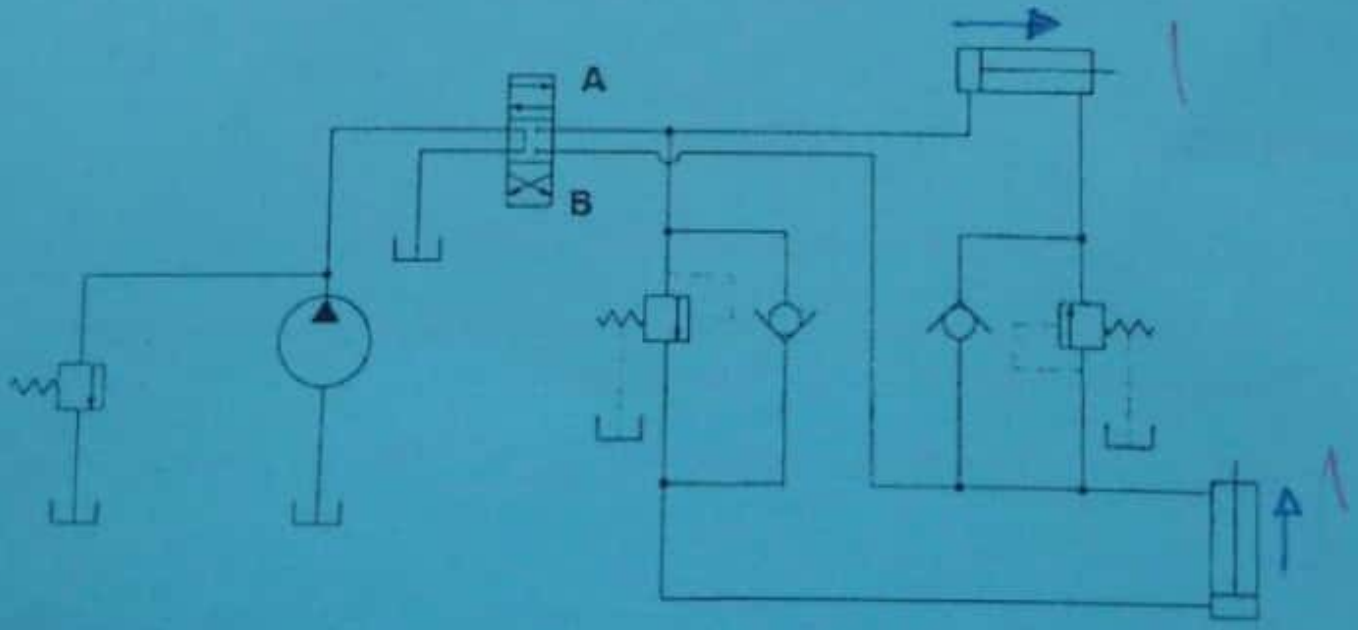


Figure 1A

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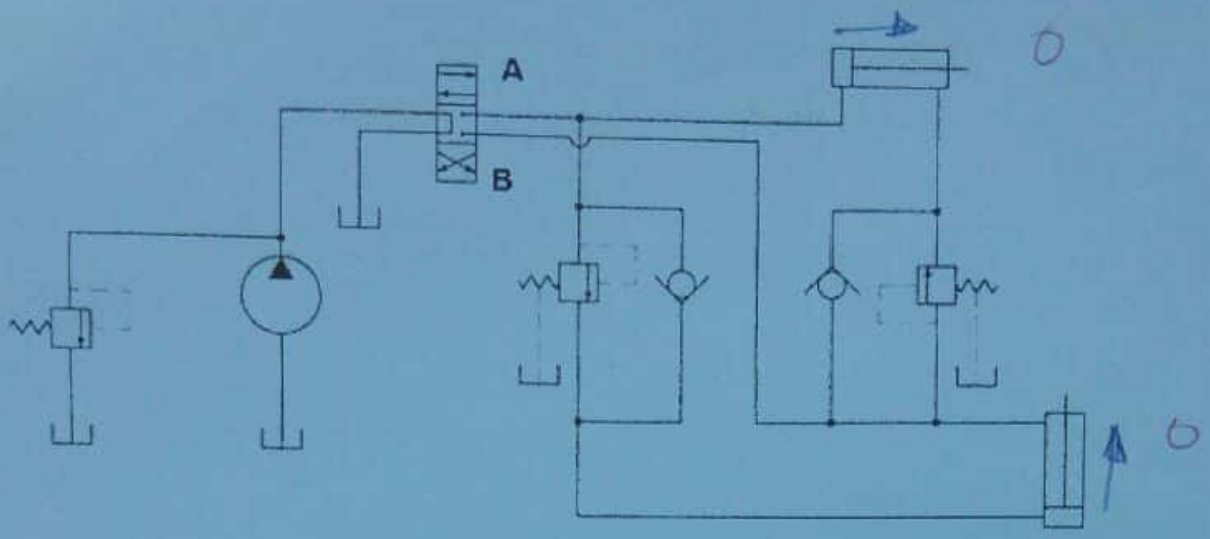


Figure 1B

[15 marks]

Question 5

(a) Figure 2 is a drawing of hydraulic circuit with 4/3 directional control valve in Views A, B and C. Using colour coding, apply the appropriate colour to the various portions of the circuit (position of the valve is indicated with shading) . Indicate with arrow the direction of the piston in each case.

(b) Produce a circuit diagram using standard symbol for the View B of Figure 2. Label the component as appropriate if this is circuit 4 of third installed system.

(c) A hydraulic Engineer designs a hydraulic system. He decided to make use of the following components: double acting cylinder, a booster, an electric motor, hydraulic pump, 4/3 directional control valve, 2/2 directional control valve (normally closed), an accumulator, pressure relief valve, a reservoir and various hose connectors.

Produce a possible circuit diagram using standard symbols with the chosen components.

[20 marks]

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5. Figure 2 is a hydraulic circuit diagram, name the various components as per label provided at respective spaces.

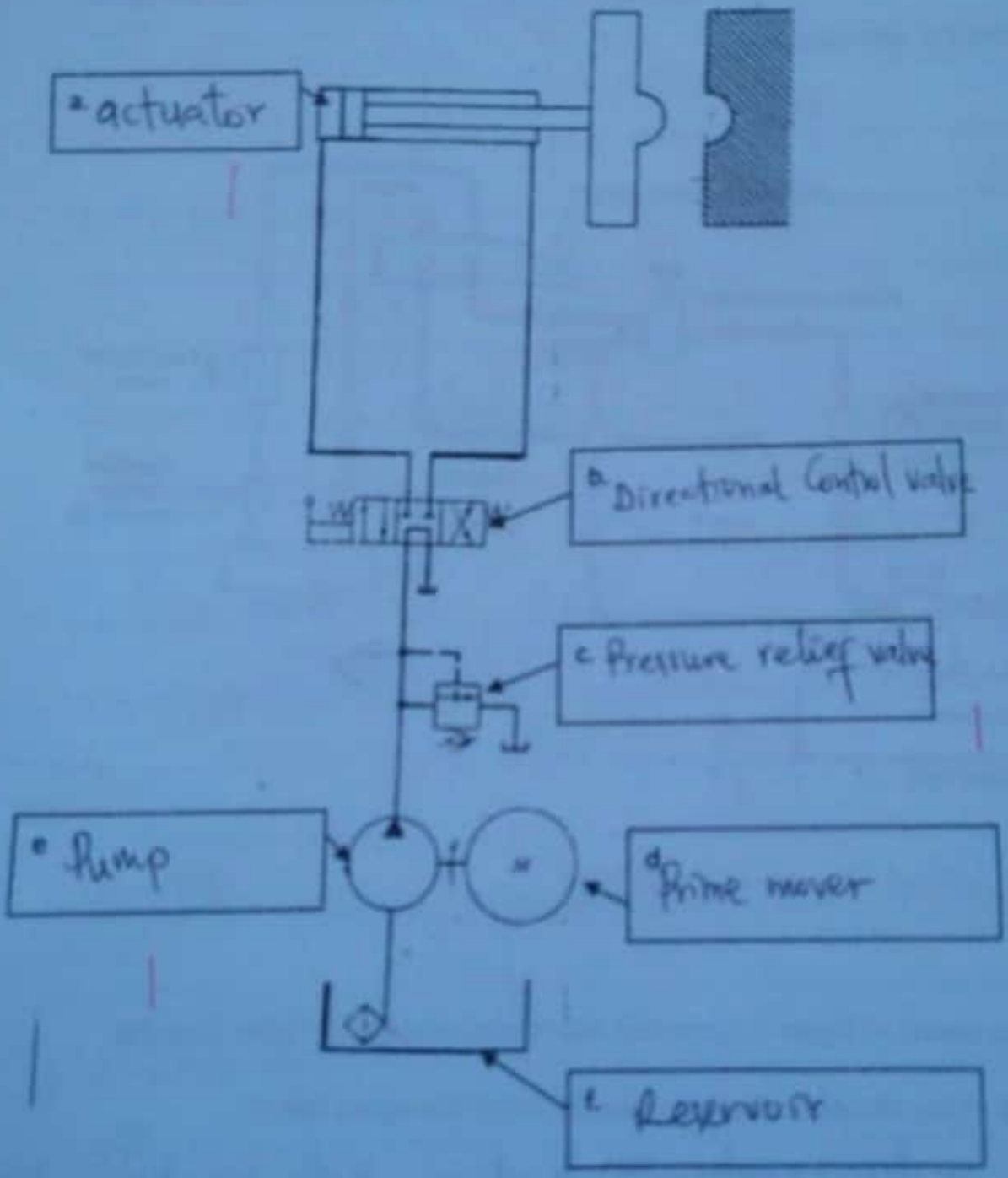
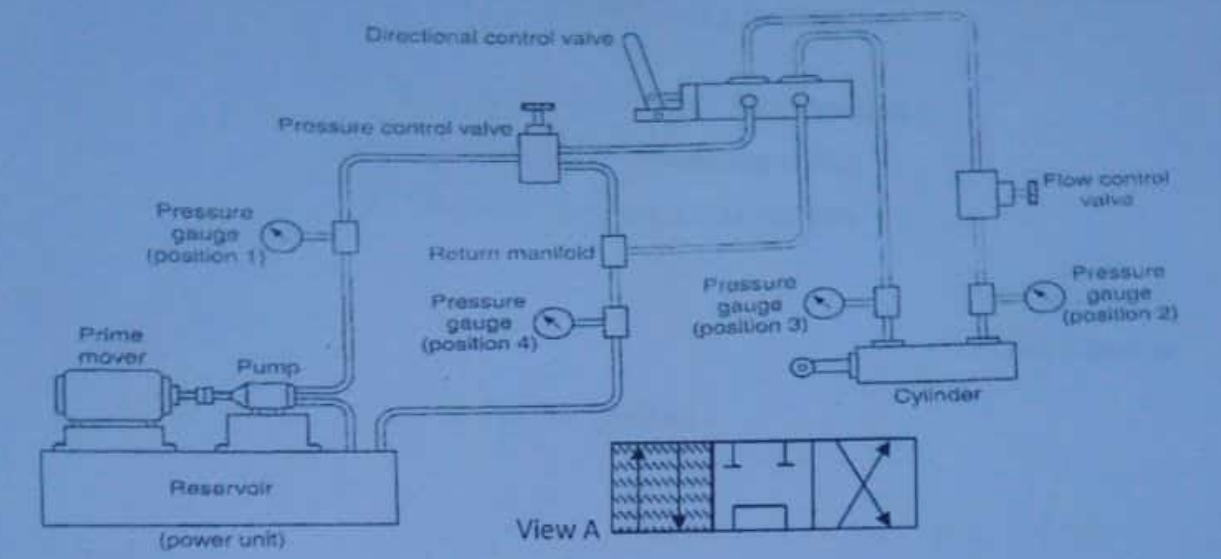
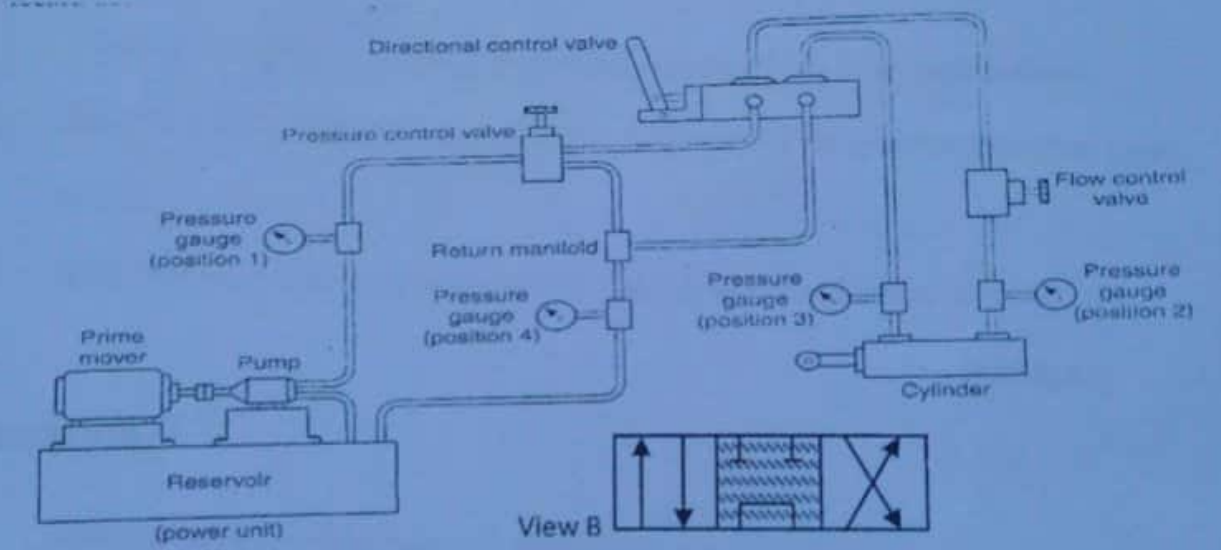


Figure 2.

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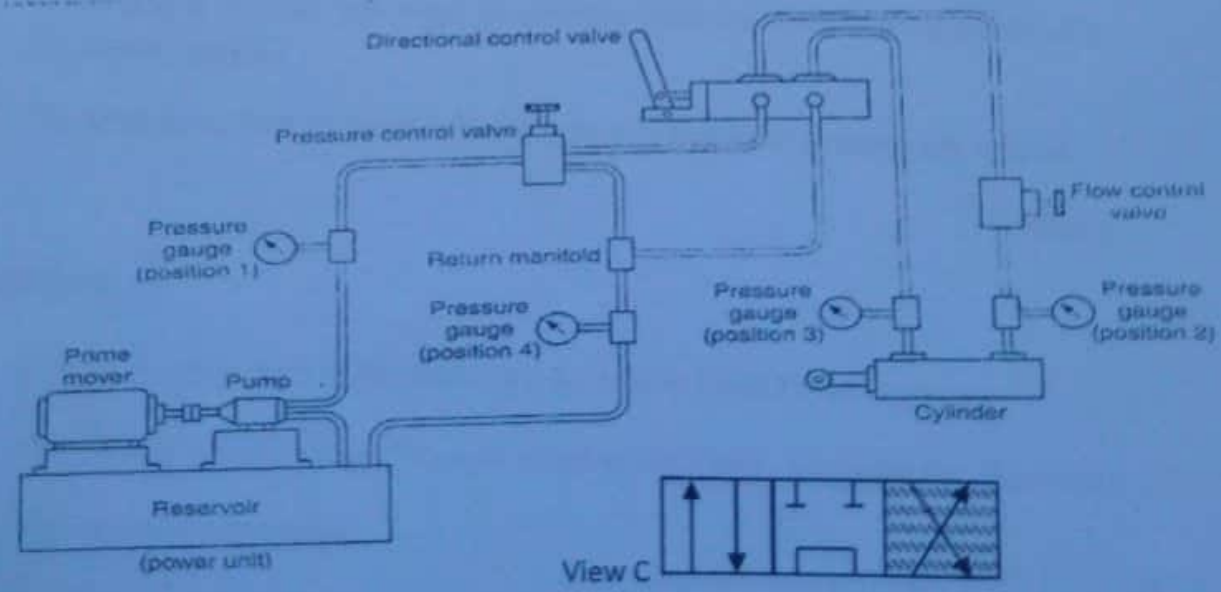
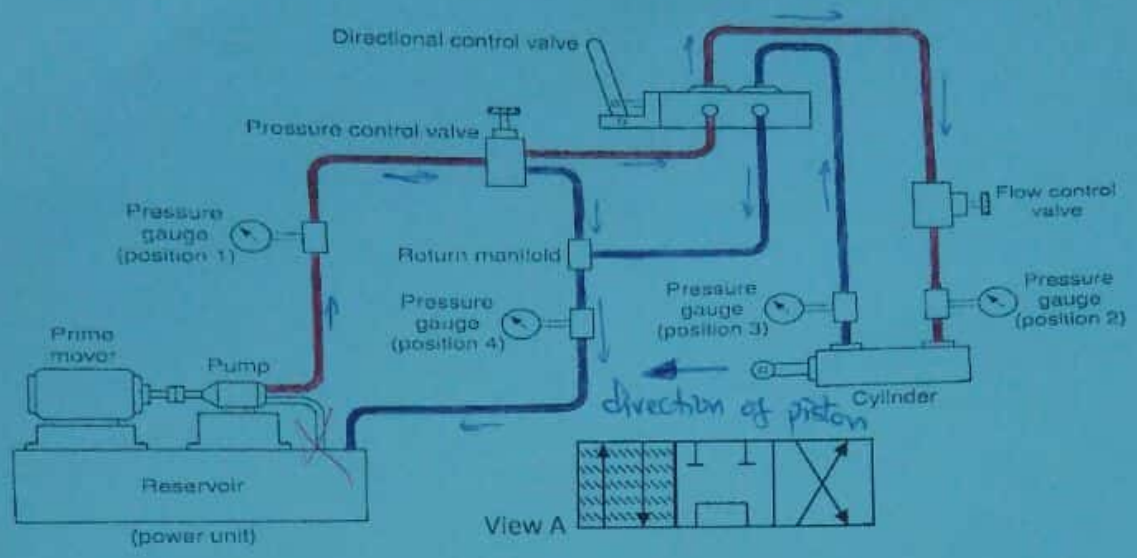
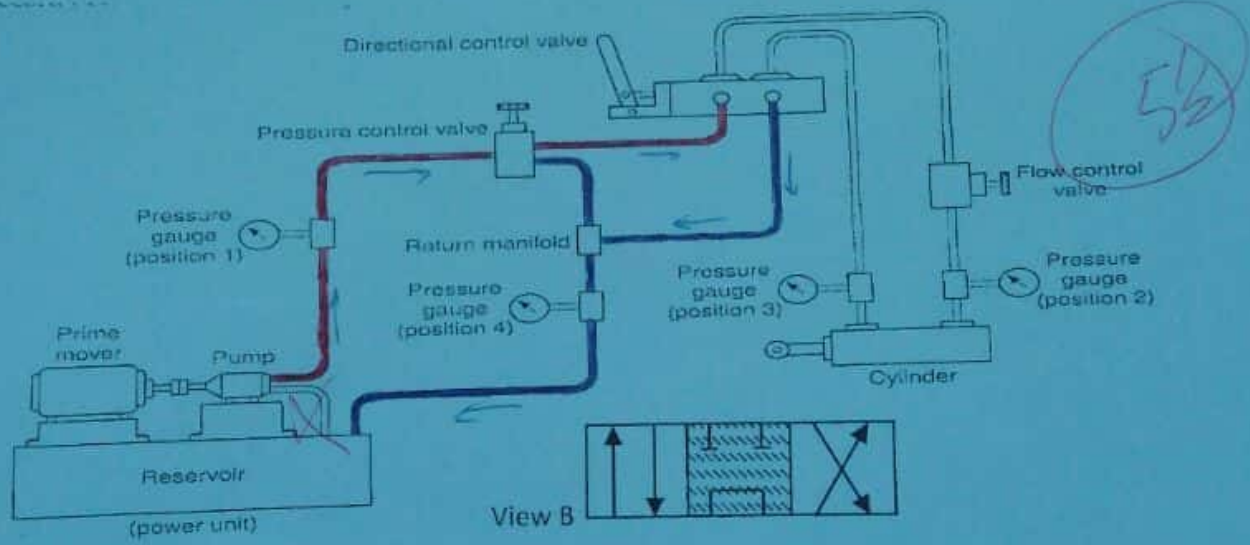


Figure 1

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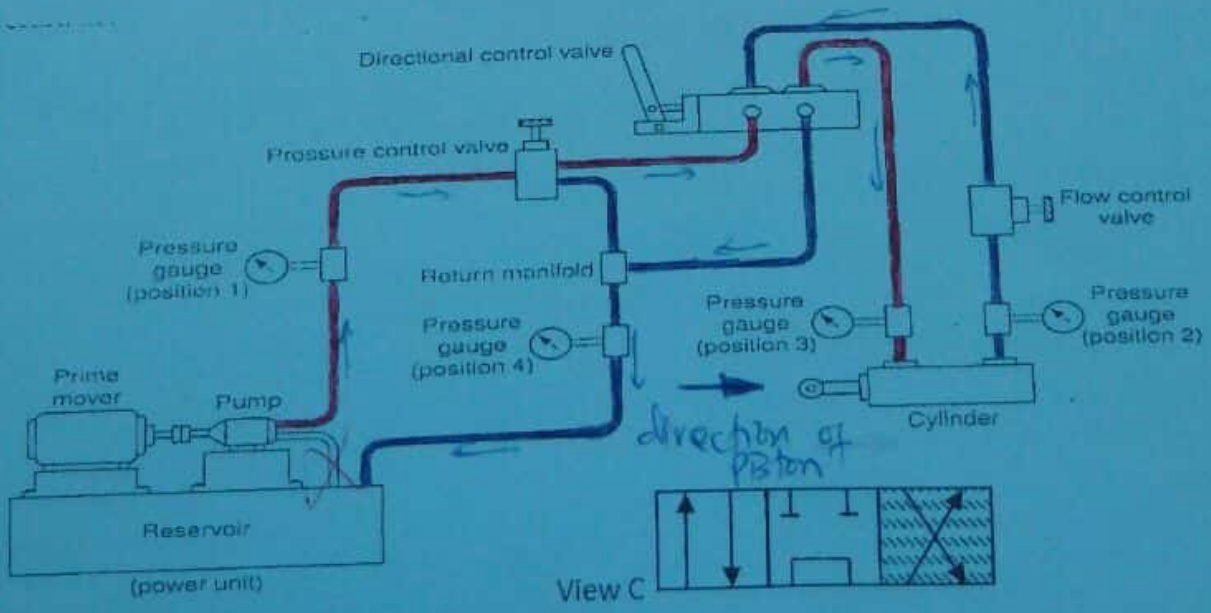


Figure 2

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2672508.

3. Figure 1 is a drawing of hydraulic circuit with four-way directional control valve. Using colour coding, apply the appropriate colour to the various portions of the circuit in the case where the actuator is retracting.

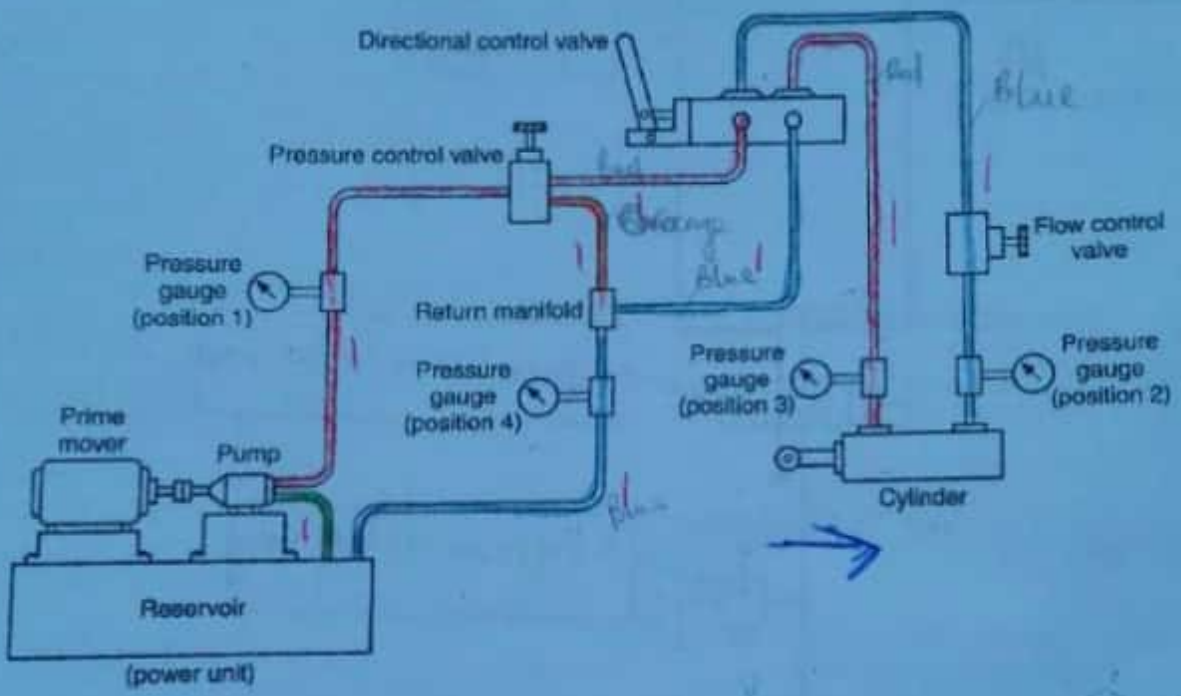


Figure 1

4. Considering a circuit of figure 1. Normally, the retraction speed is higher than the extension speed. What should be done to make the retraction speed lower?

A pressure reducing valve can be placed in the line towards retraction. A reduced pressure will reduce the speed of retraction.

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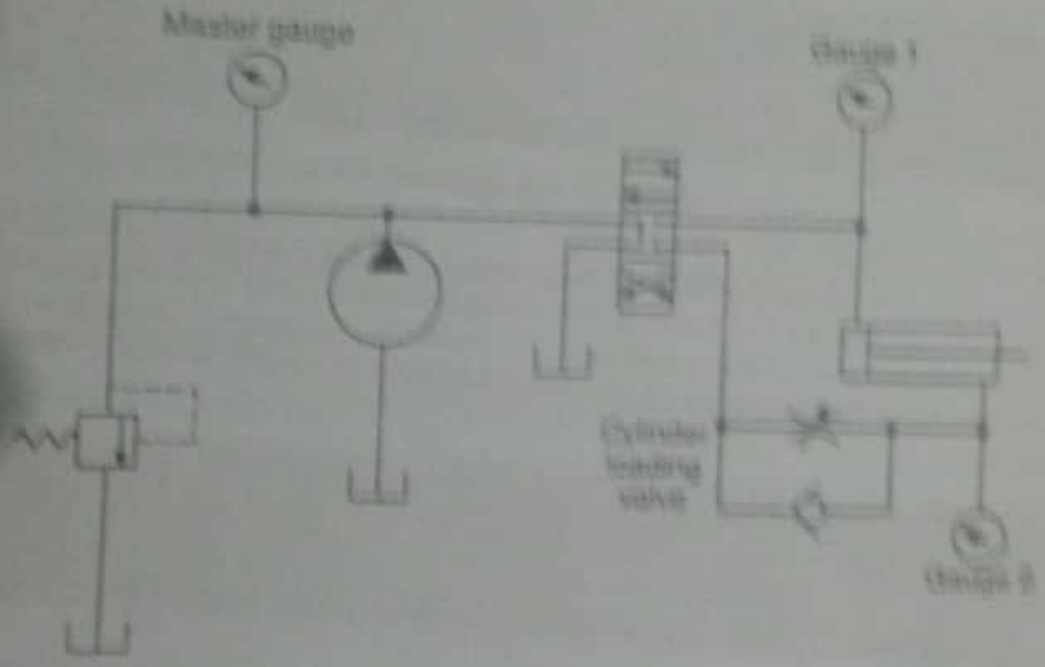
Cylinder Force and Speed

Name _____ Date _____

Cylinder force and operating speed are basic to the operation of most hydraulic circuits. Understanding the capability of a cylinder to generate force and speed is fundamental to selecting a cylinder that can efficiently operate within the requirements of the application. This activity requires the calculation of both force and speed. Then, a hydraulic test circuit is used to verify the theoretical calculations.

Activity Specifications

Calculate the theoretical force and speed that a cylinder is capable of generating during extension and retraction at a specific maximum system pressure and flow. Set up a test circuit to verify the results of the calculations and identify possible reasons for variations. The circuit diagram shown below includes the power unit and the necessary test components. The loading valve simulates a load on the cylinder by restricting flow out of the cylinder. It is used only during the force-measuring part of the activity.



Required Components

Access to the following components is needed to complete this activity:

- Hydraulic power unit (pump, reservoir, and prime mover)
- Pressure control valve (relief valve)
- Directional control valve (four way, three position, tandem center)
- Flow control valve with return check valve (used as the cylinder loading device)
- Three pressure gauges
- Cylinder (double acting, single rod)
- Manifolds (adequate number to make gauge and other connections)
- Hoses (adequate number to allow assembly of the system)

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